



A statistical approach to detect disparity prone features in a group fairness setting

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Abstract

The use of machine learning models in decision support systems with high societal impact raised concerns about unfair (disparate) results for different groups of people. When evaluating such unfair decisions, one generally relies on predefined groups that are determined by a set of features that are considered sensitive. However, such an approach is subjective and does not guarantee that these features are the only ones to be considered as sensitive nor that they entail unfair (disparate) outcomes. In this paper, we propose a preprocessing step to address the task of automatically recognizing disparity-prone features that does not require a trained model to verify unfair results. Our proposal is based on the Hilbert-Schmidt independence criterion, which measures the statistical dependence of variable distributions. We hypothesize that if the dependence between the label vector and a candidate is high for a sensitive feature, then the information provided by this feature will entail disparate performance measures between groups. Our empirical results at test our hypothesis and show that several features considered as sensitive in the literature do not necessarily entail disparate (unfair) results.

Keywords Group fairness · Disparity prone feature · Hilbert–Schmidt independence criterion · Proxy feature

1 Introduction

The use of Machine Learning (ML) methods to deal with real-world problems has grown exponentially in the last decades [1, 2]. In particular, ML is currently integrated in every decision support (DS) system with critical societal impacts. This fact has raised increasing concerns about the (un)fairness of such ‘artificial deciders’ in several contexts [3], such as in recidivism risk prediction [4], facial recognition

systems [5, 6], platforms for job applications [7], hate speech and abusive language detection [8]. This motivated several efforts towards both the detection and mitigation of unfairness in ML and DSs. The former are mostly centered in the detection of undesirable bias, while the latter seeks to reduce such biases. These methods fall into four main categories, namely, pre-processing [9–12], in-processing [13–15],

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post-processing [16–18], and hybrid-processing¹ [19–21]. For a recent survey on these methodologies and discussions on related fairness tensions, see [22].

Fairness of decision systems is commonly assessed via metrics that rely on the outcomes,² of decision models, and inspired by anti-discrimination laws under which decision policies or practices can be declared as discriminatory based on their effects on people belonging to a certain sensitive demographic groups [24], e.g., gender, race, age and sexual orientation. Still, fairness is a social construct [25] and an ethical concept [26, 27], and its definition remains prone to subjectivity.

To overcome subjectivity, several fairness notions have been recently proposed to objectively measure and to capture different aspects of fairness. These include group-based notions (e.g., [16, 28]), individual-based notions (e.g., [28, 29]), and causal and counterfactual based notions (e.g., [30, 31]). Even though these metrics would replace human subjectivity with objective metrics, they are still dependent on subjective decisions.

Indeed, most of the above metrics rely on the choice of particular features (variables) with respect to which fairness will be measured. For instance, in the case of group-based metrics, fairness is mostly measured with respect to disparate performance and decision results among different subpopulations. For example, when predicting recidivism risk [4], disparate results among the white and black subpopulations, or among men and women. In most of the fairness analysis in ML and DS problems, it is assumed that the population is divided into privileged and unprivileged groups on which disparate results are expected. In other words, the sensitive features that discriminate these groups are known a priori. Examples of commonly considered sensitive features include race, gender, age, sexual orientation, etc. See, e.g., [32] for a set of ML problems and the sensitive features frequently considered in the literature. It is worth mentioning that most of these features are defined as sensitive ones according to legal protection against discrimination. For instance, it is illegal in the US³ and in Europe⁴ to discriminate on the basis of gender, race, sexual orientation, etc. In fact, several countries in the world have implemented similar AI protection rules. However, there is no global approach.⁵

However, the choice of sensitive features also remains subjective and, as they are chosen a priori, it is dataset agnostic and does not necessarily entail outcome disparities in a given use case scenario. Indeed, consider an ML problem in which both gender and race are among the set of features. Depending on problem addressed, one may have uneven results with respect to race but similar results regardless the gender. In this situation, one may only be aware of race as a disparity-prone feature when training the ML model. As further illustration, consider one of the above mentioned platforms for job applications. Ethically, physical disabilities of candidates should not constitute a discriminating (i.e., it should be considered sensitive) of candidates. However, if the job is indeed of a physical nature, then some physical abilities may be required and, therefore, this information should not be considered as sensitive.

In this paper, we address this critical issue of identifying disparity prone feature or subfeatures (i.e., corresponding protected groups or subgroups), and we propose to defining them as features that entail unfair outcomes. However, to verify disparities among different (sub)groups, one usually needs to train the ML model and calculate the pertaining fairness metrics. This step may be computationally costly depending on the ML model is adopted. Thus this raises the question: how to detect possible disparities before the training step?

To tackle this issue, we propose a statistical approach based on the Hilbert-Schmidt independence criterion (HSIC) [33], which is commonly used in data science [34] to measure the dependence between two matrices (or vectors). The HSIC has been used as a feature selection method [35, 36], for instance, in classification or regression tasks. The HSIC is usually used under the assumption that “good features” are those that maximize the dependence degree between these features and the vector of labels or values, and one selects the subset of features with the highest dependence degrees, as they are more relevant to the output. In [37], the authors used the empirical estimate of HSIC to derive a supervised version of Principal Component Analysis (PCA). The idea was to perform dimensionality reduction while maximizing the dependence between the projected data and the vector of outcomes. HSIC has also been used when defining the loss-function for regression of classification task [38, 39]. In this context, a noteworthy application addresses fairness in automatic decisions by learning models that reduce disparities between sensitive groups [40, 41].

Our hypothesis in this work is that, for a given feature, if

1. this feature brings information that splits people into different groups, and

¹ These methods combine different fairness interventions such as in-processing and post-processing.

² Some works also focus on procedural fairness, i.e. the decision-making processes (means) that lead to the outcomes [23]. However, this concern will remain outside the scope of this paper.

³ <https://www.whitehouse.gov/ostp/ai-bill-of-rights/#applying>.

⁴ https://ec.europa.eu/newsroom/dae/document.cfm?doc_id=60419.

⁵ <https://www.taylorwessing.com/en/interface/2023/ai-are-we-getting-the-balance-between-regulation-and-innovation-right/ai-regulation-around-the-world>.

2. the HSIC⁶ between this feature and vector of labels is high,

then there will be disparate outcomes among the (sub)groups discriminated by this feature. The intuition is that the features with high HSIC are important to the classification task as they carry key information to discriminate (sub)classes. Hence, it is likely to observe disparate outcomes among the discriminated (sub)groups.

Note that HSIC (or its normalized variant, NOCCO) simply indicates the features' potential to entail disparities, and features with high HSIC should then be analyzed by practitioners and domain experts to decide whether they should be considered as sensitive, especially, when a cause-effect relation is detected. For example, in Sect. 5.3 we will observe that more "hours per week" has as an effect a higher "income". Similarly, the charge degree in recidivism risk prediction [4] may split the population between those who committed a felony or a misdemeanor, but should not be considered as a sensitive one. Therefore, although some features may achieve a high HSIC value, they should not be considered as sensitive ones if splitting people based on such a feature does not entail ethical concerns. Moreover, sensitive features may split the population into more than two groups. For instance, as race, one may have whites, blacks and Asians. For each subfeature (e.g., race_whites, race_blacks and race_Asiants), a high HSIC will indicate (1) if the feature is sensitive and (2) which group division may lead to unfair (disparate) results.

As we will see, HSIC indicates both the sensitive features and the associated groups of individuals that may lead to unfair outcomes. More precisely, high HSIC values between features (or subfeatures, in the case of categorical data) and the label vector may entail disparate results. We attest our hypothesis with empirical evaluations on several datasets frequently used in the literature, and that relate high HSIC values to high group fairness measures. This HSIC-based preprocessing approach thus constitutes a noteworthy tool for researchers and practitioners to automatically detect those features that should be considered sensitive in ML or DSs tasks. It is important to recall that, as our approach consists in a preprocessing method, one avoids the computational cost of verifying unfair results based on a trained ML model.

It is important to stress that our approach deviates from previous attempts such as [23, 24] in two major aspects, namely, by seeking an automatic approach to choosing which features should be considered as features that entail unfair outcomes, and by claiming that the choice of such

features should take into account both the task and use case scenario. In particular, we show that certain features often considered as sensitive from ethical and social standpoints do not result in outcome disparities, and thus their use in ML and DSs should not be prohibited. In fact, our approach also enables proxy detection as we will discuss in Sect. 5.

The rest of this paper is organized as follows. In Sect. 2, we outline the notations used in this paper. Section 3 discusses the Hilbert-Schmidt independence criterion and its normalized version. The proposed approach to automatically detect disparity-prone features and subfeatures is presented in Sect. 4. In Sect. 5, we conduct the numerical experiments and discuss the obtained results. Finally, in Sect. 6, we present our conclusions and future perspectives.

Main contributions

Here, we simply highlight the main contributions of the paper.

- We address the problem of detecting features that entail unfair outcomes in group fairness settings, and propose a statistical approach based on the HSIC that does not require a trained model to verify disparate results.
- We present preliminary empirical results on four well-known datasets that support our hypothesis. The higher is the HSIC-based dependence measure between a feature and the labels, the higher are the disparate results entailed by using the information provided by such a feature.
- The analysis of our results also shows that several features taken as sensitive in the literature do not necessarily entail disparate (unfair) results. This is useful for practitioners when building fair algorithms. More effort can be devoted to mitigating biases towards sensitive features that are actually associated with disparity outcomes.
- We also raise some perspectives for future work. Our proposed approach can be easily extended to decide whether a combination features (e.g., the use of features describing gender and race) can lead to disparate results. Moreover, we will investigate how to adapt our framework to multiclass classification problems and to individual fairness settings.

2 Notation

Throughout this paper, we assume an underlying binary classification problem such that each m -dimensional sample is assigned to a class $y \in \{-1, 1\}$. The set of n samples and the associated vector of classes are represented by $\mathbf{X}$ and $\mathbf{y}$, respectively. Each column of $\mathbf{X}$, defined by $\mathbf{X}^{(j)}$, describes a feature G_j , $j = 1, \dots, m$.

Very often some features in ML problems are categorical. However, as most of the learning algorithms require

⁶ In fact, as will be further discussed in this paper, we adopted a normalized version of HSIC.

numbers as input features, we consider in this paper the one-hot encoding method to transform the categorical features into binary ones. Mathematically, a vector $\mathbf{X}^{(j)}$ describing a categorical feature G_j with feature space $\{G_{j,1}, G_{j,2}, \dots, G_{j,q}\}$ can be replaced by⁷ q binary vectors $\mathbf{X}^{(j,k)}$, $k = 1, \dots, q$, that indicate which category each sample belongs to. We illustrate this scenario with the 3-dimensional dataset

$$\mathbf{X} = \begin{bmatrix} \text{male} & 2 & 30 - 60 \\ \text{female} & 0 & < 30 \\ \text{female} & 1 & > 60 \end{bmatrix}, \quad (1)$$

whose features are

G_1 : gender with possible values male or female,

G_2 : number ncrimes of committed crimes in the last 5 years, and

G_3 : age that can be less than 30, between 30 and 60 or greater than 60 years old.

After encoding the categorical features, one achieves the extended 5-dimensional dataset

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad (2)$$

where $G_{1,1} = \text{gender}_{\text{male}}$, $G_{1,2} = \text{gender}_{\text{female}}$, $G_{3,1} = \text{age}_{\leq 30}$, $G_{3,2} = \text{age}_{30-60}$ and $G_{3,3} = \text{age}_{\geq 60}$ are the novel subfeatures. Note that, after conducting one-hot encoding for all categorical features, one increases the dataset dimension from m to $\tilde{m}$.

3 Hilbert–Schmidt independence criterion

There are several measures adopted in feature selection tasks to evaluate the relation between inputs and outputs [42]. As examples, one may cite the correlation [43] and mutual information [44] between variables. Methods based on correlation are generally straightforward to be deployed as this measure is easy to be calculated. However, correlation only provides second-order information between variables and, therefore, does not necessarily imply dependence. On the other hand, mutual information brings the knowledge about the dependence degree between variables. The price to be paid here is that, to calculate the mutual information, one

needs to estimate the probability density function of such variables. This task is very impractical in most cases.

Another measure that has been used in recent works in the literature is the Hilbert–Schmidt independence criterion [34]. The HSIC measures the dependence degree between finite number of observations⁸ (x_i, y_i) , $i = 1, \dots, n$. Assume $\mathbf{K}_x$ and $\mathbf{K}_y$ as the kernel matrices of $\mathbf{x}$ and $\mathbf{y}$, respectively. In our analysis, we consider the radial basis function (RBF) kernel, defined by

$$K^{\text{RBF}}(z_i, z_{i'}) = e^{-\frac{1}{n}(z_i - z_{i'})^2},$$

and the linear kernel, defined by

$$K^{\text{linear}}(z_i, z_{i'}) = z_i z_{i'}.$$

However, other kernels can also be considered (see [45] for details). An empirical calculation of HSIC is given by

$$\text{HSIC}(\mathbf{x}, \mathbf{y}) = \frac{\text{tr}(\mathbf{K}_x \mathbf{H} \mathbf{K}_y \mathbf{H})}{(n-1)^2},$$

where $\mathbf{H} = \mathbf{I} - n^{-1} \mathbf{e} \mathbf{e}^T$ is the centering matrix and $\mathbf{e}$ is a n -dimensional column vector of ones. Note that, even if the dataset $\mathbf{x}$ is centered, this is not necessarily true for the kernel $\mathbf{K}_x$. To centralize this kernel, one applies the centering matrix $\mathbf{H}$ on both sides of $\mathbf{K}_x$. Recall that

$$\text{tr}(\mathbf{K}_x \mathbf{H} \mathbf{K}_y \mathbf{H}) = \text{tr}(\mathbf{H} \mathbf{K}_x \mathbf{H} \mathbf{K}_y) = \text{tr}(\tilde{\mathbf{K}}_x \mathbf{K}_y),$$

where $\tilde{\mathbf{K}}_x$ is the centered kernel. Moreover, when we multiply $\mathbf{K}_x$ on the left (resp. right) by $\mathbf{H}$, one removes its columns (resp. rows) mean.

As highlighted in [41], the HSIC calculation is sensitive to the scale of the observations and, therefore, an appropriate normalization should be conducted to compare relative dependence degrees. We thus consider a normalized version of HSIC called NOCCO (NORmalized Cross-Covariance Operator) [46]. It is defined as follows:

$$\text{NOCCO}(\mathbf{x}, \mathbf{y}) = \text{tr}(\mathbf{R}_x \mathbf{R}_y),$$

where

- $\mathbf{R}_x = \mathbf{H} \mathbf{K}_x \mathbf{H} (\mathbf{H} \mathbf{K}_x \mathbf{H} + n\epsilon \mathbf{I}_n)^{-1}$,
- $\mathbf{R}_y = \mathbf{H} \mathbf{K}_y \mathbf{H} (\mathbf{H} \mathbf{K}_y \mathbf{H} + n\epsilon \mathbf{I}_n)^{-1}$,
- ϵ is a regularization parameter (e.g., 10^{-6}), and
- $\mathbf{I}_n$ is a $n \times n$ identity matrix.

It is important to highlight that the use of NOCCO has two main advantages. Firstly, as the calculation is based on the

⁷ One generally replaces by $q-1$ binary features to avoid a redundant column. However, in our analysis, the use of either q or $q-1$ binary features lead to the same results.

⁸ We here consider $\mathbf{x} = \{x_i\}_{i=1}^n$ and $\mathbf{y} = \{y_i\}_{i=1}^n$ as vectors. However, HSIC can be also used to calculate the dependence degree between matrices.

trace of matrix products, it is easily calculated (we do not need to estimate, for instance, probability density functions). Secondly, the use of kernels brings more information than second-order moment to measure the relation between variables. Therefore, it is useful to estimate the dependence degree between them.

4 Proposed approach

The purpose of this work is to automatically detect disparity-prone features and the associated impacted groups. Our hypothesis is that if a feature that provides information that splits groups of people (e.g., if the individual is white or black) is important to increase the performance of a machine learning model (measure by means of NOCCO), then this feature may create disparities between such groups in terms of performance measures.

Consider the extended dataset $\tilde{\mathbf{X}}$, with columns $\tilde{\mathbf{X}}^{(1)}, \dots, \tilde{\mathbf{X}}^{(m)}$, obtained after encoding categorical features. As mentioned in Sect. 3, HSIC can be used to measure the dependence between two vectors. For each numerical feature and encoded features (i.e., for all columns of $\tilde{\mathbf{X}}$), we calculate NOCCO with respect to the vector of outcomes $\mathbf{y}$. For numerical features, the dependence degree with respect to the vector of labels is the NOCCO itself. In the case of categorical features, if this feature is a binary one, then the NOCCO for both groups will be the same and, therefore, we only need to calculate it once. Otherwise, if the categorical features have three or more categories, we assume as the dependence degree the maximum NOCCO among the associated encoded features. This procedure leads to m measures of dependence

$$d_j = \text{tr}(\mathbf{R}_{G_j} \mathbf{R}_y), \quad j = 1, \dots, m, \quad (3)$$

where $\mathbf{R}_{G_j} = \mathbf{H} \mathbf{K}_{G_j} \mathbf{H} (\mathbf{H} \mathbf{K}_{G_j} \mathbf{H} + n \mathbf{I}_n)^{-1}$ and $\mathbf{K}_{G_j}$ is the kernel matrix associated with feature G_j . In the example of Sect. 2, if NOCCO for $age_{\leq 30}$ is greater than the NOCCO for both age_{30-60} and $age_{\geq 60}$, we define d_3 as the dependence measure between $\tilde{\mathbf{X}}^{(4)}$ and $\mathbf{y}$.

Based on all d_j , $j = 1, \dots, m$, we may evaluate whether a feature should be considered as a disparity-prone feature. In this paper, we assume an automatic procedure that takes into account a predefined threshold t . Algorithm 1 presents a pseudo-code of our proposal. For each feature G_j , if $d_j \geq t$, G_j is considered as a feature that entail unfair outcomes and the $G_{j,k'}$ that leads to the higher NOCCO is the harmed group.⁹ Otherwise, we do not consider G_j as a disparity-prone

feature. In our empirical setting, we take t to be the median value among all $d_1, \dots, d_m$. Note that the decision-maker can adopt another strategy to define t (even without a threshold, which leads to a more subjective choice).

5 Empirical evaluation

In this section, we present the experimental setup to evaluate our proposal on several real datasets. We identify possible disparity prone features as well as the associated harmed groups. Moreover, we provide a comparison between the obtained NOCCO values and the fairness measures with respect to those features with high NOCCO values.

5.1 Datasets

To assess our proposal, we consider four datasets frequently used in the literature: Adult income,¹⁰ COMPAS recidivism risk [4], Law School Admission Council (LSAC) [47] and Taiwanese credit default [48]. A brief description and how we preprocessed some features are provided in the sequel. See [32] for further details as well as the list of features frequently considered as sensitive ones in the literature.

- **Adult income dataset:** In this dataset, the aim is to predict whether a person makes over 50K a year based on the following features: age, workclass, educational-num (educational degree), marital status, occupation, relationship (husband, not in family, other relative, own child, unmarried or wife), race (Indian-Eskimo, Asian-Pacific Islander, black, white or other), gender (male or female), capital gain, capital loss, hours per week and native-country. After removing samples with missing values, we achieved 45,222 samples. We also rearranged some categorical features: age = {< 25, 25-60, > 60}, workclass = {private, non-private}, marital-status = {married, never-married, other}, occupation = {office, heavy-work, service, other} and native-country = {US, non-US}. One typically assumes age, race and gender as sensitive features.
- **COMPAS recidivism risk dataset:** This dataset has the goal of classifying individuals as potential criminal recidivist. There are 6167 samples and 8 features, namely sex (male or female), age_cat (age category - less than 25, between 25 and 45 or greater than 45), race, juv_fel_count (number of juvenile felony), juv_misd_count

⁹ Recall that our proposal indicates which feature (G_j) and which subgroup ($G_{j,k'}$) can lead to disparate results. For example, NOCCO may indicate that nationality is a candidate for a disparity-prone fea-

Footnote 9 (Continued)

ture. If there are subgroups within ethnic groups, a high NOCCO may observe disparities in Africans vs. non-Africans but not in Europeans vs. non-Europeans.

¹⁰ <https://archive.ics.uci.edu/ml/datasets/adult>.

Algorithm 1 (Automatic approach for sensitive features detection)

Input: $\mathbf{X}$ and $\mathbf{y}$.
Output: Set of disparity prone features $\mathcal{S} = \{G_{j'}, G_{j''}, \dots\}$ and harmed groups $\mathcal{S}^g = \{G_{j',k'}, G_{j'',k''}, \dots\}$.
1: **Encoding categorical features:** Encode the categorical features and obtain the extended dataset $\tilde{\mathbf{X}} \leftarrow \text{encode}(\mathbf{X})$.
2: **Calculate the kernel matrix of $\mathbf{y}$:** $\mathbf{K}_y = \text{kernel}(\mathbf{y})$
3: **Calculate the NOCCO for all features and subfeatures:**
for $j \in \{1, \dots, m\}$ **do**
 if G_j is either a numerical or binary feature **then**
 Calculate the kernel matrix associated with G_j : $\mathbf{K}_{G_j} = \text{kernel}(G_j)$
 Calculate the dependence measure: $d_j = \text{tr}(\mathbf{R}_{G_j} \mathbf{R}_y)$
 else
 Calculate the maximum NOCCO for the categorical variable:
 for $k \in \{1, \dots, q\}$ **do**
 Calculate the kernel matrix associated with $G_{j,k}$: $\mathbf{K}_{G_{j,k}} = \text{kernel}(G_{j,k})$
 Calculate the dependence measure: $d_j^k = \text{tr}(\mathbf{R}_{G_{j,k}} \mathbf{R}_y)$
 end for
 Define the dependence measure for feature G_j : $d_j \leftarrow \max(d_j^1, \dots, d_j^q)$
 end if
end for
4: **Create the set of disparity prone features and harmed groups:** $\mathcal{S} = \emptyset$ and $\mathcal{S}^g = \emptyset$
5: **Calculate the median value among all d_j , $j = 1, \dots, m$:** $t = \text{median}(d_1, \dots, d_m)$
for $j \in \{1, \dots, m\}$ **do**
 if G_j is candidate for a disparity prone feature and $d_j \geq t$: **then**
 Update the set of disparity prone features: $\mathcal{S} \leftarrow \{\mathcal{S}, G_j\}$
 Update the set of harmed groups: $\mathcal{S}_g \leftarrow \{\mathcal{S}_g, G_{j,k^*}\}$ such that $d_j^{k^*} = \max(d_j^1, \dots, d_j^q)$
 end if
end for

(number of juvenile misdemeanor), juv_other_count (number of others infractions), priors_count (number of priors) and c_charge_degree (charge degree - felony or misdemeanor). We here rearranged race {Caucasian, African-American, Other}. Race and sex are assumed as sensitive features.

- **Law School Admission Council (LSAC) dataset:** In this dataset, there are 23,726 candidates described by 11 features: decile1b (decile given the grades in the 1st year), decile3 (decile given the grades in the 3rd year), lsat (score), ugpa (undergraduate GPA), zfygpa (1st-year law school GPA), zgpa (cumulative law school GPA), fulltime (full-time or part-time work), fam_inc (family income), male (whether the student is male or female), race and tier (tier of the law school). The goal is to predict whether a candidate would pass the bar exam. One assumes gender and race as sensitive features.
- **Taiwanese credit default dataset:** In this dataset, the aim is to predict customers' default payments in a Taiwanese institution. The 23 features are the following: limit_bal (amount of given credit), sex (male or female), education (graduate school, university, high school or others),

marriage (married, single or others), age (less than 35 or at least 35), repayment status (six features, from April to September), amount of bill statement (six features, from April to September) and amount paid (six features, from April to September). We assumed sex, education and marriage as sensitive features.

5.2 Fairness metrics

To attest our hypothesis that the NOCCO value can be associated with fairness measures (and, hence, used as a preprocessing approach to detect features that entail unfair outcomes), we compare the obtained values of fairness measures with respect to features with high NOCCO values. These measures are based on the following performances extracted from the trained ML model:

- True positive (TP): # of instances correctly classified as class 1.
- True negative (TN): # of instances correctly classified as class -1.

- False positive (FP): # of instances wrongly classified as class 1.
- False negative (FN): # of instances wrongly classified as class -1.

Aiming at evaluating some fairness measures when splitting the dataset into different groups of people, we restrict the aforementioned performance measures to such groups. Suppose, for example, that feature G_j contains information that splits people into groups $\{G_{j,1}, \dots, G_{j,q}\}$. As the NOCCO for group $G_{j,k}$ will indicate the dependence degree between such a group and the vector of labels, when evaluating fairness, we consider the disparities when people is divided between those belonging to group $G_{j,k}$ and those that does not. Therefore, we restrict the performance measures such that $FP_{G_{j,k}}$ means the number of instances belonging to group $G_{j,k}$ wrongly classified as class 1 and $TN_{-G_{j,k}}$ is the number of instances not belonging to group $G_{j,k}$ correctly classified as class -1 (we referred to $-G_{j,k}$ as the instances not belonging to group $G_{j,k}$).

Given the performance measures conditioned on groups, we may define the fairness measures considered in our experiments:

- Predictive equality (PE): A classifier satisfies predictive equality if both groups have equal false positive rate. In other words, we should have low values of

$$f_{PE} = \left| \frac{FP_{G_{j,k}}}{FP_{G_{j,k}} + TN_{G_{j,k}}} - \frac{FP_{-G_{j,k}}}{FP_{-G_{j,k}} + TN_{-G_{j,k}}} \right|.$$

- Equal opportunity (EP): A classifier satisfies equal opportunity if both groups have equal true positive rate. In other words, we should have low values of

$$f_{EP} = \left| \frac{TP_{G_{j,k}}}{TP_{G_{j,k}} + FN_{G_{j,k}}} - \frac{TP_{-G_{j,k}}}{TP_{-G_{j,k}} + FN_{-G_{j,k}}} \right|.$$

- Equalized odds (EO): A classifier satisfies equalized odds if both groups have equal true positive and false positive rates. In other words, we should have low values of

$$f_{EO} = \left| \frac{TP_{G_{j,k}}}{TP_{G_{j,k}} + FN_{G_{j,k}}} - \frac{TP_{-G_{j,k}}}{TP_{-G_{j,k}} + FN_{-G_{j,k}}} \right| + \left| \frac{FP_{G_{j,k}}}{FP_{G_{j,k}} + TN_{G_{j,k}}} - \frac{FP_{-G_{j,k}}}{FP_{-G_{j,k}} + TN_{-G_{j,k}}} \right|.$$

- Overall accuracy equality (OAE): A classifier satisfies overall accuracy equality if both groups have equal correct classification for both classes -1 and 1. In other words, we should have low values of

$$f_{OAE} = \left| \frac{TP_{G_{j,k}} + TN_{G_{j,k}}}{n_{G_{j,k}}} - \frac{TP_{-G_{j,k}} + TN_{-G_{j,k}}}{n_{-G_{j,k}}} \right|,$$

where $n_{G_{j,k}}$ and $n_{-G_{j,k}}$ are the number of instances that belong and do not belong to group $G_{j,k}$.

5.3 Result analysis

In our experiments, we first calculated the NOCCO for each feature and encoded feature. Thereafter, we calculate the median NOCCO and define the features that could be considered as disparity prone ones. Aiming at attesting the usefulness of NOCCO as a measure to detect features that entail disparate results, we compare the NOCCO values with a set of fairness measures. For this purpose, we trained a Random Forest¹¹ classifier using k-fold cross validation (with 10 folds). For each fold and encoded features, we calculated performances and fairness measures (equal opportunity, predictive parity, equalized odds and overall accuracy equality, by taking the absolute difference). For categorical features with more than two groups, we calculated these measures by assuming a one-vs-all strategy.

We present results obtained for the Adult income, COMPAS, LSAC and Taiwanese credit default datasets in Figs. 1, 2, 3 and 4, respectively. The dashed lines indicate the trend between NOCCO and the fairness metrics. In NOCCO calculation, we assumed the RBF kernel. With respect to the computational cost, the average time of our experiments, for each dataset, was approximately 15 min.¹²

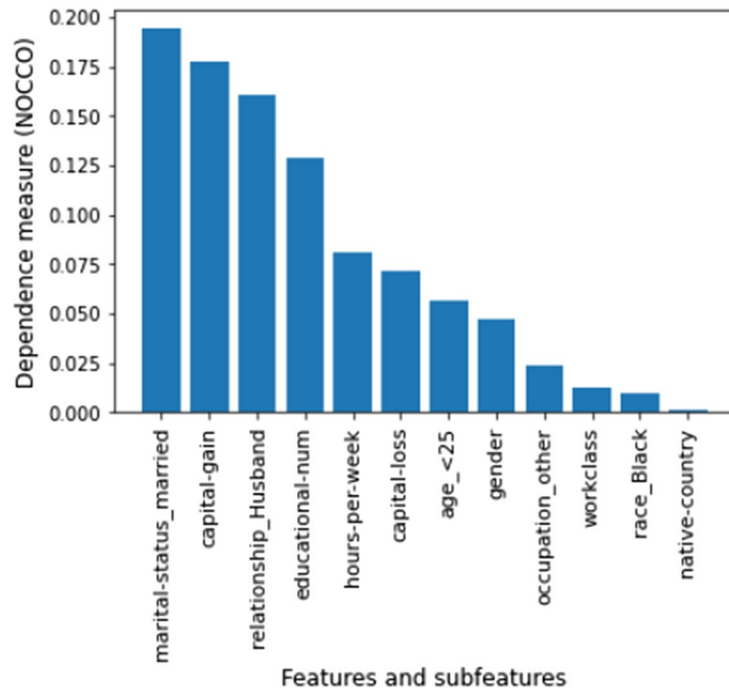
Note that, for all datasets and all fairness measures (see Figs. 1b, 2b, 3b, 4b), there is a statistically positive relation between the NOCCO and the fairness metrics (surely, with positive and negative deviations from the increasing trend line). This suggests that the higher is the NOCCO, the higher is the fairness measures (i.e., the higher are the unfair results).

Results on the adult income dataset In the results with the Adult income dataset, we obtained a median NOCCO equals to 0.0642. Features whose NOCCO is greater than this threshold are marital status, capital gain, relationship, educational num, hours per week and capital loss (see Fig. 1a). As education-num indicates the individual's educational degree, it is likely that higher the educational degree (i.e., higher the qualifications for well-paid jobs) higher the obtained income. Similarly, higher incomes can be justified by people whose hours per week worked are high. Therefore,

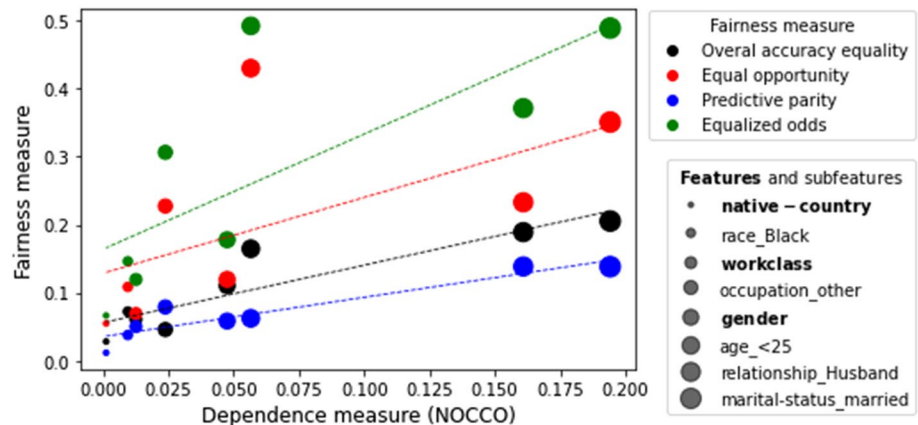
¹¹ Although in our experiments we used a Random Forest classifier, proposed approach can be applied to any classification model. Therefore, any classifier could be used to assess the relation between the NOCCO values and the fairness measures.

¹² Computation performed on a laptop Intel Core i7-8565U, CPU 1.80 GHz, 8.00 GB RAM, Python 3.9.

Fig. 1 Results for the Adult income dataset (with RBF kernel). In the legend, features appear in boldface, whereas subfeatures appear in regular font



(a) Dependence measure (NOCCO).

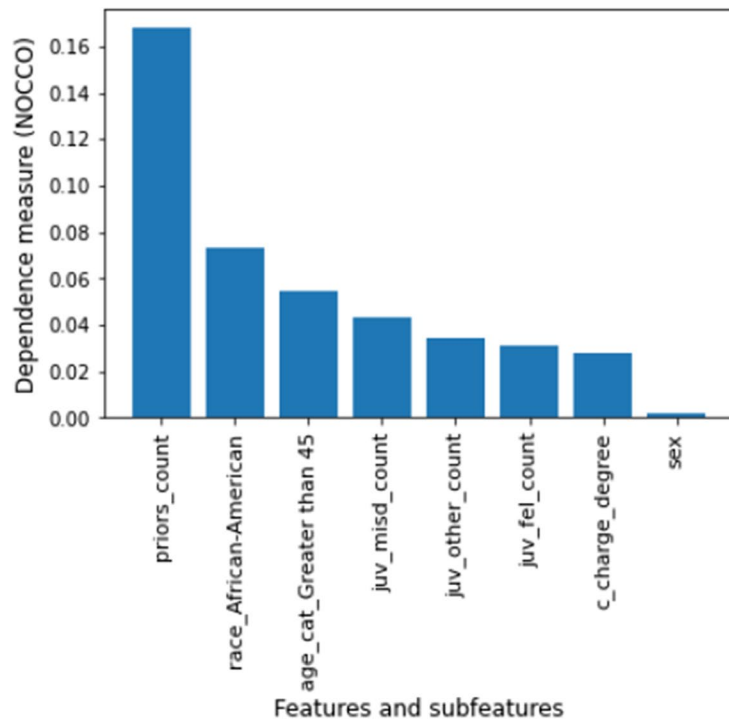


(b) Relation between NOCCO and fairness measures.

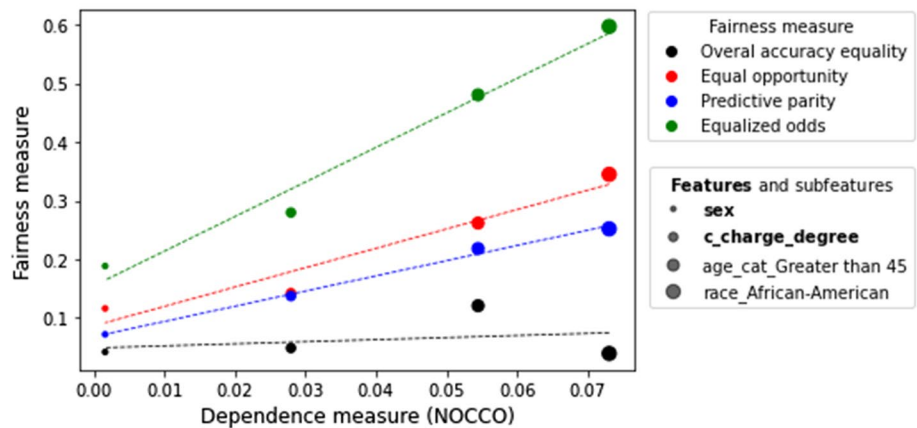
as either education-num or hours per week can be directly associated with income, and splitting people based on these features does not create unfair concerns, these should not be considered as sensitive ones. However, marital status and relationship are disparity prone ones. Although gender is frequently pointed as a feature that entail disparate results in this dataset, marital status and relationship appeared as higher candidates. The higher NOCCO was obtained when splitting people between married and non-married. Moreover, NOCCO is high when we divide the instances in those who are a husband in a relation and those who are not. As in this dataset, husband in a relation is a proxy to know that the person is a man, this information is directly associated with gender. Therefore, not only gender should be consider

as a sensitive feature when looking at men/women, but we should also be aware of the information provided by relationship. From Fig. 1b, we also observe that, although there is a clear increasing trend between NOCCO values and the fairness metrics, some low values of NOCCO were associated with high values of fairness metrics. Surely, it is difficult to define a deterministic relation between these measures and, therefore, deviations from the trend line may appear. Another interesting aspect of this dataset is that race led to a very small NOCCO. This suggested that, although frequently considered as a sensitive feature, in this problem, it will not lead to disparate results. We attested this conclusion in the fairness measures presented in Fig. 1b. The fairness measures are very small when splitting people by race.

Fig. 2 Results for the COMPAS dataset (with RBF kernel). In the legend, features appear in boldface, whereas subfeatures appear in regular font



(a) Dependence measure (NOCCO).



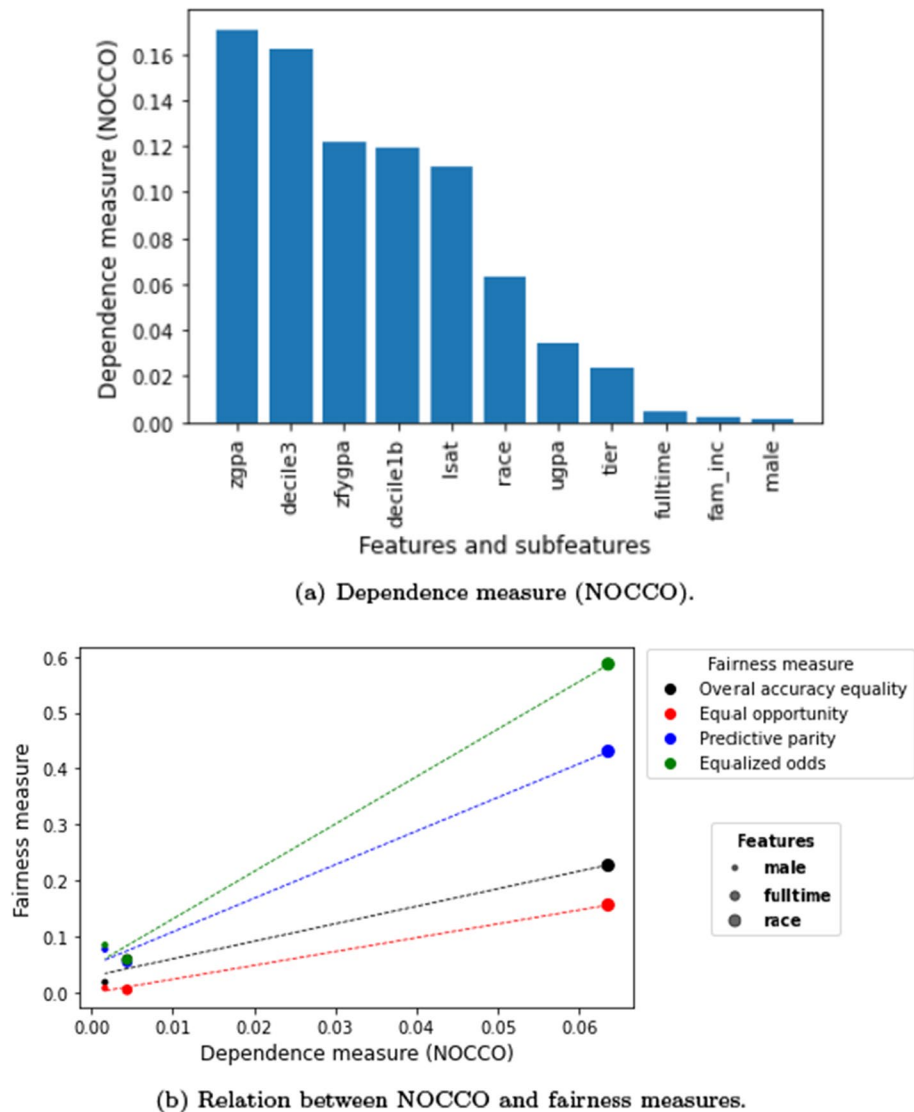
(b) Relation between NOCCO and fairness measures.

Results on the COMPAS dataset In COMPAS dataset, the literature assumes gender and race as sensitive features. In the NOCCO values presented in Fig. 2a and with $t = 0.0390$, only race (when dividing people between African-American and other ones) and age category (greater than 45) could be considered as features that entail unfair results. Although sex is frequently considered as sensitive, the obtained NOCCO value indicates that, in this dataset, it does not entail disparities between male and female. See in Fig. 2b that fairness measures when splitting people by sex are much lower in comparison with race or age category. Another interesting

remark presented in Fig. 2b is that the relation between NOCCO and most of the fairness measures are nearly linear.

Results on the LSAC dataset The NOCCO values for the LSAC dataset are presented in Fig. 3a. Only race and sex is considered as candidates for sensitive features in this dataset. With a median value of 0.0635, race should be considered as a disparity-prone feature and male should not. Indeed, as can be attested in Fig. 3b, while splitting people by race creates unfair results (we obtained high values of fairness measures), a division by gender does not (the fairness measures are very small).

Fig. 3 Results for the LSAC dataset (with RBF kernel)

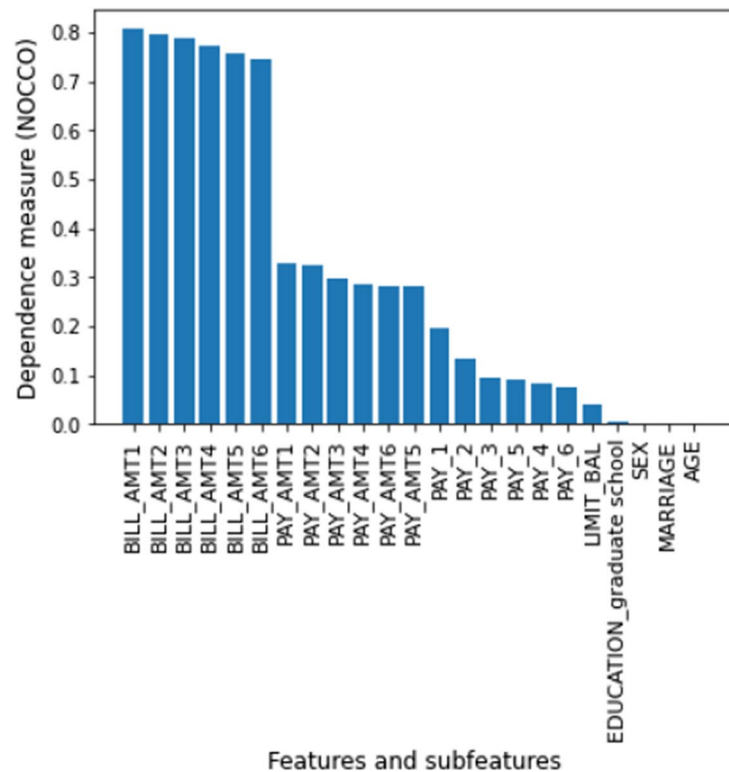


Results the Taiwanese credit default dataset Another interesting result was obtained with the Taiwanese credit default dataset. One may see in Fig. 4a that features generally considered as sensitive ones (sex, education and marriage) have very small NOCCO values. Features associated with payment status have a much higher dependence on class labels than the considered sensitive features. Surely, this dependence is justified (and without ethical concerns) by the natural correlation relation between payment status and credit concession. Therefore, with a median NOCCO value of 0.2789, no feature could be defined as disparity-prone ones. This result is attested by the small values (with most of them lower than 0.10) of the fairness metrics presented in Fig. 4. Indeed, if one compares these fairness metrics with the ones achieved with the previous datasets, one may assume that they are small.

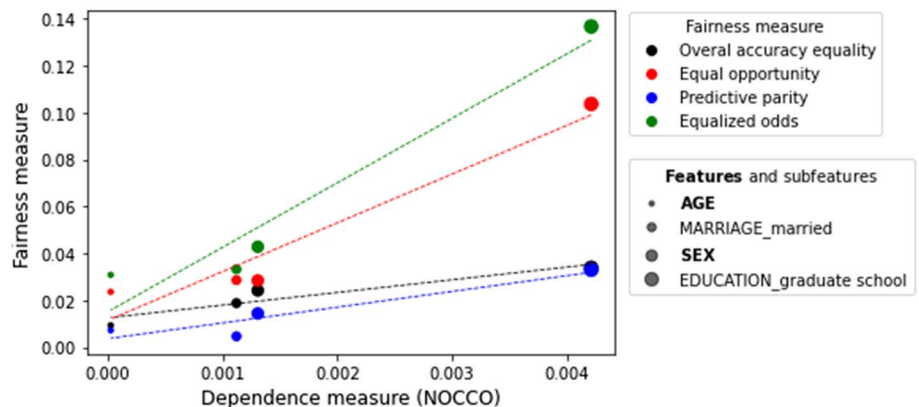
5.4 Consistency with linear kernel

To verify the consistency of the previous results by assuming another kernel, we evaluate the relation between linear kernel-based NOCCO and the fairness measures. The results for the Adult income, COMPAS, LSAC and Taiwanese credit default datasets are presented in Figs. 5a–d, respectively. Clearly, all the results with the linear kernel are consistent with the ones obtained by using the RBF kernel. The only difference can be seen in Adult income dataset with respect to the detected harmed group for race. Instead of race_White (as detected by using the RBF kernel), we here assigned race_Black as the harmed group. The reason for this disparity lies in the number of samples for each race category. As the number of samples whose categories are different from white or black is less than 5%, when using one-vs-all strategy, the results for race_White and race_Black are practically the same. Therefore, we

Fig. 4 Results for the Taiwanese credit default dataset (with RBF kernel). In the legend, features appear in boldface, whereas subfeatures appear in regular font



(a) Dependence measure (NOCCO).



(b) Relation between NOCCO and fairness measures.

can safely consider that there was not a disparity between the obtained results.

6 Conclusions and perspectives

In this paper, we proposed an automatic approach to detect disparity prone features based on a normalized version of the Hilbert-Schmidt independence criterion. As our method only requires the dataset and labels, we are able to detect features that entail unfair outcomes without the need of training a machine learning model and evaluating

the performance measures. Numerical experiments on several datasets and with different kernels attested that there is an increasing trend between the NOCCO values and the fairness metrics. Our findings indicate that the higher is the dependence between a sensitive feature and the vector of labels, the higher are the outcome disparities with respect to groups discriminated by such a feature. In other words, a low NOCCO value indicates that the (categorical) feature and the vector of outcomes are practically independent. In this case, the information provided by such a feature has no effect on the predicted outcome and it is unlikely to be a source of bias. Therefore, the statistical dependence calculated from

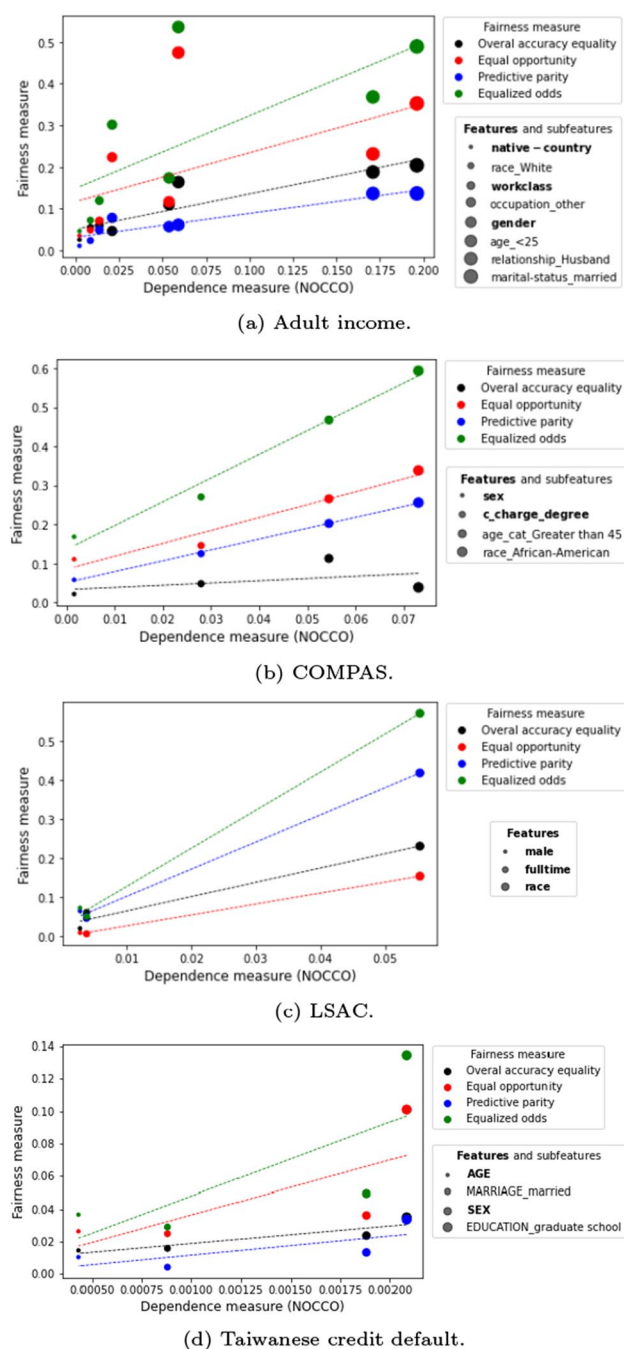


Fig. 5 Relation between NOCCO (with linear kernel) and fairness measures for all datasets. We compare **a** (adult income), **b** (COMPAS), **c** (LSAC), **d** (Taiwanese credit default) with Figs. 1b, 2b, 3b and 4b, respectively. In the legend, features appear in boldface, whereas subfeatures appear in regular font

the dataset and labels proved to be a measure that can guide researchers and practitioners in detecting disparity-prone features, especially in situations and case scenarios where the pertaining fairness notions seem hard to identify.

To the best of our knowledge, this is the first attempt to automate the detection of disparity-prone features in a

dataset (before the training step). This is a problem that has not been properly addressed in the literature, and there is a true need to tackle it within algorithmic fairness given its potential to promote more equitable outcomes in algorithmic decision making. Therefore, we highlight that the proposed approach has a clear impact in Explainable Artificial Intelligence (XAI) and it constitutes a valuable tool in cases where sensitive features are not known. Moreover, we showed that features frequently considered as sensitive in the literature may not entail disparate results. These findings are thus useful when developing a machine learning model that takes into account information from predetermined disparity-prone features to mitigate disparate results provided by them. However, our approach should be used with particular attention as it does not guarantee all features with low NOCCO will not entail high disparities. In addition, it is not meant to detect salient/protected features. This thorough analysis should be carried out by domain experts.

This work opens interesting perspectives for future works. Although we used the NOCCO to calculate the dependence degree between labels and vectors of features that describe groups in the population (e.g., if the person is black or not), this measure can also be applied to coalitions of features (e.g., subgroups). This asks for the dependence degree between two or more features (such as race and gender) and the vector of outcomes. This may show that combining information provided by two or more features may increase the chance of achieving disparate results. Note that the combination of features can be achieved by computation of the dependence degree between a matrix and the vector of outcomes, which can be handled by the NOCCO value. As another perspective, we intend to extend the proposed approach to deal with multiclass classification problems. In such scenarios, for each available class, we may detect which features may entail unfair outcomes. Finally, future work could evaluate the robustness of our proposal for different thresholds, and according to the level of severity required by the use case and the decision support system.

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Data availability All codes and datasets are freely available at https://github.com/GuilhermePelegri/HSIC_sensitive_features.git.

Declarations

Conflict of interest The authors have no conflict of interest concerning this research.

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